David Jeehwan Lee

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Dual Citizenship (U.S. & Republic of Korea)

Education

Korea Advanced Institute of Science and Technology (KAIST) Daejeon, South Korea
B.S. in Electrical Engineering, Cum Laude (GPA: 3.82/4.30) *Mar. 2022 – Feb. 2027*

Daegu IL Science High School Daegu, South Korea
High School Diploma, Valedictorian *Mar. 2019 – Feb. 2022*

Experience

IMPACT Lab, KAIST Jan. 2026 – Present
Undergraduate Researcher (Advisor: Prof. Minkyu Je) *Daejeon, South Korea*

Quad-channel time-domain capacitance-to-digital converter (B.S. thesis)

- Designed a CDC that reads four capacitive sensors at once: each sits in one cell of a single quadrature-phase swing-boosted RC oscillator, so all four convert every period without multiplexing.
- Used ISF-based simulations to show that uneven rise and fall times turn slow flicker noise into timing jitter, which led me to a low-jitter relaxation oscillator and flicker-aware gated-ring-oscillator TDCs.
- Achieved 0.102 fF resolution in 6 μs at 110 μW in 0.18-μm, 0.9-V CMOS simulations, estimating droplet velocities with 0.22% error versus 10.6% for a time-multiplexed readout.

Flexible ion-gel pressure-sensor array readout (collaboration with IOEL, KAIST)

- Built an automated readout for a 10×10 ion-gel capacitive pressure-sensor array: multiplexers route each of the 100 sensors to a PCap04 CDC under Arduino control (I^2C), with a Python host for calibration and display.
- LCR sweeps showed the gel's high resistance needs a slow measurement, so slowing the CDC discharge doubled the pressure response; with wiring parasitics calibrated out, it showed a live pressure map at 7.7 frames/s.

Tearney Lab, Massachusetts General Hospital, Harvard-MIT HST Jun. 2026 – Aug. 2026
Research Intern (Advisor: Prof. Guillermo J. Tearney) *Boston, USA*

- Developed a 360° scanning μOCT probe for real-time guidance in deep brain stimulation (DBS): the fiber core stays still and only an elliptical mirror spins, avoiding the image distortion and light loss of a rotating fiber.
- Designed the optics in Zemax OpticStudio, fabricated the fiber core (cleaving, splicing, and tapering single- and multi-mode fiber), and integrated a Nanoscribe 3D-printed mirror into a handheld motorized prototype.
- Validated a 6–8 μm focus (about a tenth of a hair's width) over ∼200 μm extended-depth-of-field

AI-PIM Semiconductor Design Research Center, KAIST Oct. 2025 – Dec. 2025
Research Intern (Funded by the Ministry of Science and ICT) *Daejeon, South Korea*

- Ported an ASIC HyperRAM controller to an Artix-7 FPGA to ease a RISC-V AI SoC's memory-bandwidth bottleneck, resolving AXI4-to-HyperBus clock-domain crossings and tri-state I/O constraints; validated with ILA.
- Integrated SoC peripherals and verified microSD access, DMA transfers, and interrupts with bare-metal C tests.

Asian Science and Technology Pioneering Institutes (ASPIRE) League Jun. 2025
KAIST Student Representative, Undergraduate Engineering Design Challenge *Tokyo, Japan*

- Represented KAIST at the Institute of Science Tokyo with a full participation grant.
- Employed design thinking methodologies to prototype upcycled bamboo DIY kits for the Tanabata festival.

Capital Defense Command, Republic of Korea Army Aug. 2023 – Feb. 2025
Military Service *Seoul, South Korea*

- Completed mandatory service as an army sergeant, serving as a squad leader and a combat fitness instructor.

Samsung Electronics Feb. 2023
Shining Star Program Trainee *Hwaseong, South Korea*

- Trained in the 8 core semiconductor processes; co-designed a wearable IC device for the hearing-impaired.

Publications & Presentations

[1] **D. J. Lee**, W. Youn, and M. Je, "A Quad-Channel Simultaneous-Sampling CDC Based on a Quadrature-Phase Swing-Boosted RC Oscillator," *IEIE Fall Conference 2026, Undergraduate Paper Competition*, Nov. 2026.

[2] **D. J. Lee**, A. Ortega, H. Schulz-Hildebrandt, B. Davis, and G. J. Tearney, "Circumferential Scanning μOCT Probe for Real-Time Guidance in Deep Brain Stimulation," Poster, *HST-Wellman Summer Institute for Biomedical Optics*, Boston, MA, Aug. 2026.

Technical Projects

LSTM Breath Detection for a Self-Powered CO_2 Sensor Node | *Deep Learning* Jun. – Aug. 2025

- Ported a lab NDIR capnography chain (lock-in, temperature-compensated Beer–Lambert, end-tidal detection) from MATLAB to Python and built a simulator that turns 279 recorded exhalations into the node's 2 Hz CO_2 stream.
- Trained a 5,025-parameter LSTM that detected 87% of held-out walk-by exhales at 0.8 false alarms/h vs. 78% at 2.2/h for the best rule-based detector, with the gain on weak 120–250 ppm breaths (92% vs. 68%).
- Showed the model held F1 at 0.87 in an unseen 2,000 ppm room and would cost only 0.2–0.5% of the node's awake power, so inference can run on the sensor instead of a host.

TinyML Wake-Word Detection on a Raspberry Pi | *On-Device AI* Sep. 2025 – Dec. 2025

- Built an end-to-end wake-word pipeline (voice activity detection, MFCC, CNN) at 92.7% validation accuracy and tuned real-time streaming detection to a 90% hit rate with few false triggers.
- Found that INT8 quantization cut the model size by ∼40% but stalled the speedup on the CPU (unpacking overhead), showing that software compression alone hits a wall without matching hardware.

Leadership & Service

Reporter, EE Newsletter, KAIST School of Electrical Engineering Mar. 2025 – Present
English Mentor, KAIST Science Outreach Program (KSOP) Sep. 2025 – Jun. 2026
Korean Buddy, KAIST International Scholar and Student Services Mar. 2025 – Jun. 2025
Korean Student Representative (fully funded), KOFST Young Generation Forum Jul. 2023
Student Volunteer, Scientists and Engineers Without Borders, Myanmar Jan. 2018

Honors & Awards

Undergraduate Paper Award, The Institute of Electronics and Information Engineers (IEIE) 2026
Special Warrior Award, Republic of Korea Army (ROKA) 2024
KAIST Science Gifted Student Scholarship, KAIST 2021
Dong-gu Education Foundation Scholarship, Daegu Metropolitan City 2021

Skills

Hardware: Embedded Systems (Raspberry Pi, Arduino), Digital SoC Design (Xilinx Vivado, Verilog), Analog Circuit Design (Cadence Virtuoso), Optical Design (Zemax OpticStudio)
Software: ML (PyTorch, TensorFlow), C/C++, Python, MATLAB
Certificate: Computer Utilization, Korea Chamber of Commerce and Industry
Languages: English (Native; TOEFL 118/120, TOEIC 990/990, OPIc AL), Korean (Native), French (A2)